

An AI-Driven Academic Collaboration Platform With Intelligent Skill-Growth Matchmaking and Startup-Support Chatbot

Abstract— This application is about Dispersed communication methods like emails, messaging apps, department-specific portals, and manual alerts frequently hinder academic collaboration in colleges. Particularly for students interested in skill development or startup-oriented growth, these disjointed processes result in restricted access to mentorship, difficulty creating teams, and poor visibility of prospects. The intelligent matchmaking engine at the heart of Connect-X examines user activities, interests, interactions, and skill profiles to comprehend individual learning styles and suggest tailored actions. This entails recommending pertinent projects, mentors, groups, events, skill-building exercises, and educational materials that support the student's development in their chosen field. Furthermore, a context-aware recommendation system presents appropriate options at the appropriate moment, including funding programs, research calls, hackathons, and seminars. Additionally, Connect-X incorporates an AI chatbot assistant that provides direction, mentor recommendations, collaboration links, and educational materials to aid students in articulating, honing, and working together on their company ideas. Connect-X facilitates better access to mentorship, improves academic coordination, and enables students to develop their ideas and talents from the first notion to the final academic or entrepreneurial result through this cohesive and intelligent ecosystem.

Keywords— *AI-based matchmaking, role-based access management, contextual recommendation engine, web-based academic center, business idea support, and chatbot assistant.*

I. INTRODUCTION

Strong learning environments, innovation promotion, and student growth all depend on productive academic collaboration. Nonetheless, the majority of higher education departments continue to rely on disjointed lines of communication, including classroom announcements, emails, messaging apps, separate department websites, and printed notices. Students find it challenging to find pertinent opportunities, establish connections with mentors, or work together with peers who have similar interests because of the information silos created by this dispersed environment. As a result of a lack of awareness, assistance, or prompt direction, numerous great academic or entrepreneurial ideas get unrealized.

There is a growing trend toward intelligent educational systems that can comprehend student demands and offer tailored help as a result of the quick advances in artificial intelligence. Conventional departmental portals usually just function as information boards; they are unable to identify skill gaps, assess user behavior, analyze interests, or suggest activities in advance. Students frequently miss deadlines, have unequal access to resources, and have ineffective departmental collaboration since they have to manually look for events, mentors, or partners.

Its AI-powered matchmaking system, which examines user behavior, academic interests, skill profiles, and interaction patterns, is a crucial part of Connect-X. The system knows what a user is interested in and what talents they are looking to develop, rather than just assigning them to tasks. Connect-X offers tailored recommendations based on this comprehension, including suggested projects, mentors, groups, workshops, and educational opportunities. This aids students in following a well-defined course of action to improve their abilities, engage in pertinent departmental activities, and work with peers more successfully.

Additionally, Connect-X offers a safe idea workspace where instructors and students may work together on documents, discuss proposals, monitor versions, and restrict permissions using role-based access control. This allows for teamwork and criticism while protecting early-stage academic or business work. By automatically exposing pertinent departmental notices, seminars, conferences, research calls, scholarships, and other opportunities, a context-aware recommendation engine enhances the system. It provides a unified platform that lowers communication gaps, boosts visibility of opportunities, and supports the growth of both academic and startup-oriented students by combining AI-driven matchmaking and with integrated chatbot assistance.

II. LITERATURE SURVEY

Pambudi et al. [1] present a conference study on the “digital revolution of startup matchmaking,” framing how AI and computer-science techniques can algorithmically connect startup founders with the right collaborators, mentors, and resources within innovation ecosystems. Their work (ICIC 2023) situates matchmaking as a data-driven pipeline—collecting user attributes, inferring compatibilities, and operationalizing matches to accelerate venture formation and knowledge exchange.

Jyotshna et.al [2] proposed a **career guidance platform** combining a conversational AI chatbot with an automated **resume generator**. The system learns from user inputs—academic background, interests, skills, aspirations—and provides personalized **career suggestions**, **skill-gap analyses**, and **actionable guidance**; the resume generator operationalizes that advice into job-ready artifacts. Reported results highlight user acceptance and outline a three-tier architecture with room for future integration of job market data and advanced recommenders.

Sitanggang et.al [3] master's thesis uses the Double Diamond service-design approach and multi-method qualitative research to investigate the needs of older women in Finland and Indonesia for an AI-powered matchmaking platform. The following are important findings for inclusive, human-centered AI platforms: privacy attitudes, trust, device usage habits (mobile + WhatsApp), and accessibility hurdles (vision, tech anxiety). For meaningful social connection and decreased loneliness, the study suggests culturally sensitive design, user-friendly interactions, and unambiguous privacy promises.

Faqihi and Miah [4] presented an conceptual framework for AI-driven Talent Management Systems (TMS) that combines Diffusion of Innovation (DOI) theory with Technology-Organization-Environment (TOE) through design science. Their design artefact is an intelligent HRM automation system with a talent intelligence module that connects evaluation, planning, and growth pathways. They also highlight dangers, including as bias, transparency, corporate adoption obstacles, and SME limits. In order to apply AI in worker contexts safely and effectively, the article emphasizes structured machine learning techniques and governance.

While emphasizing ethical issues—privacy, bias, integrity, and the digital divide—Shah's [5] book (Wiley, 2023) summarizes how AI may assist personalized learning routes, administrative automation, assessment/feedback, and student involvement. It offers useful methods for teachers to incorporate AI, such as generative models and chatbots.

AI can enhance teamwork while maintaining a significant role for human contact, according to recent conference and journal results. In their evaluation of AI-enhanced collaborative platforms, Islam et al. [6] make the case for striking a balance between intelligent recommendations and peer-to-peer learning mechanisms (live sessions, chats, incentives) in order to improve community engagement and avoid an over-reliance on automation. This strategy is similar to Connect-X's design of activity-aware suggestions combined with human mentorship and group work.

According to the Esangbedo et.al [7] and the ten-year systematic review (SSHO, 2025), if tasks are well-designed and emotional engagement is encouraged, AI tools (ML, NLP, recommenders) can improve group performance, knowledge building, and social presence. This has implications for Connect-X to pair matchmaking with inclusive task structures and facilitation prompts.

Kovari et.al [8] designed an user-friendly processes and unambiguous privacy assurances are recommended by human-centered research on AI matchmaking (elderly women in Finland and Indonesia), which emphasizes trust, privacy attitudes, and accessibility as important adoption drivers across demographics. These principles are also applicable to student-faculty ecosystems. According to state-of-the-art overviews of AI in higher education.

Scutelnicu, Liviu-Andrei, and Marius Ciprian Ceobanu. [9] discussed about the contextual recommendations for events, calls, and cross-department/industry projects are supported

by evidence from structural equation modeling with primary data (n=230) that AI has a significant positive direct effect on educational opportunities, with curriculum development acting as a strong positive mediator.

Shai Kumar R et.al [10] implemented the integration of operational streams (RFID, fingerprint, and vision) with academic systems to provide dependable participation indicators is demonstrated by smart-campus attendance and CV/biometric systems. When used appropriately (aggregate and privacy-preserving), these signals can prioritize or time recommendations without storing sensitive identifiers in collaboration layers. Campus context (time, location, load) can enhance suggestion freshness and diversity, according to broader smart-campus literature on IoT-driven optimization, provided that stringent data-minimization and retention constraints are in place.

Sundarambal et al. [11] presented an AI-based departmental management platform designed to unify academic workflows and create a student-centric digital ecosystem. The system integrates study materials, automated attendance tracking, and digital notice boards into a centralized interface to streamline departmental operations. An AI-driven doubt-clarification module provides students with instant academic support, while additional modules deliver career guidance, internship information, placement data, and feedback mechanisms.

Umamaheswari et al. [12] researched about the work addresses about the inefficiencies in traditional Department Management Systems (DMS), which rely heavily on manual academic processes such as scheduling, attendance tracking, and evaluation. The authors propose a machine-learning-driven intelligent system that automates core departmental activities and enhances decision-making. By integrating predictive analytics and optimization models.

Lai, Qiujia [13] explored how artificial intelligence can refine college student education management by improving administrative efficiency, personalization, and predictive decision-making. The study introduces AGTO-ResRNN, a hybrid deep-learning model enhanced with the Artificial Gorilla Troops Optimizer, to analyze academic records, attendance logs, and student engagement metrics.

Shai Kumar R et al. [14] researched about an gesture-based mobile learning platform aimed at improving educational accessibility for deaf and mute learners. The system combines sign-language instructional videos, gesture detection, visual aids, and interactive assessments to support inclusive education. A lightweight machine-learning model interprets basic gestures and maps them to words or sentences, enhancing communication and comprehension.

Vijayaraj et al. [15] presented a paper which proposes a federated hybrid AI-powered cybersecurity framework designed for adaptive detection of sophisticated cyber threats. Conventional intrusion detection systems face challenges such as high false positives and poor adaptability; the proposed model addresses these limitations using a federated learning architecture combined with hybrid AI techniques.

The framework integrates distributed model updates, behavioral analytics, and multi-layer threat analysis to enhance detection accuracy while preserving data privacy.

III. EXISTING SYSTEM

Departmental cooperation, communication, and student support procedures at the majority of academic institutions are dependent on antiquated, disjointed, and manually operated systems. For academic coordination, students and professors must rely on numerous disjointed channels due to these systems' lack of intelligence and lack of integration. The following lists the main drawbacks of the current environment.

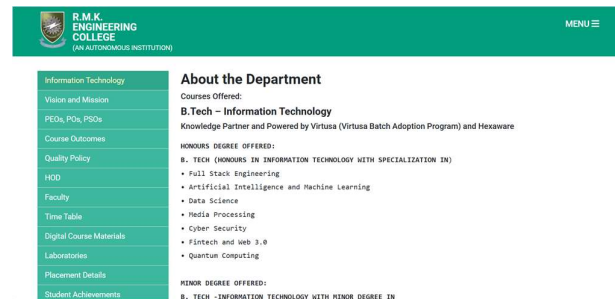


Fig 1. Existing Department management system

A. Fragmented Communication Channels

To disseminate critical updates, departments usually use a combination of printed notifications, classroom announcements, WhatsApp groups, emails, and faculty communications. As in Fig 1. Students miss to search and get updates so they frequently miss deadlines, opportunities, seminars, and academic notifications due to the disarray and irregularity of these channels. All communications cannot be viewed in one location on a single platform.

B. No Intelligent Matching or Skill-Based Recommendations

Students' interests, academic records, abilities, and involvement patterns are not examined by current methods. Consequently:

- Finding appropriate project partners, mentors, research opportunities, or groups is a challenge for students.
- Students who are good fits for departmental initiatives, events, or research cannot be found by faculty.
- Based on their academic behavior, there is no AI-based system that advises students on how to develop their talents or what activities to engage in.

C. Lack of a Secure Space for Idea Development

Students that have startup or early-stage project ideas lack due to the inability of keeping an secure setting in which they can create, revise, or save their proposals. A way to regulate who can see or comment on their thoughts. Resources for safe departmental or cross-departmental collaboration As a result,

kids are less likely to be creative and to keep their ideas to themselves.

D. No Centralized Departmental Resources

Isolated systems are usually used to store resources including lab data, instructor profiles, study material repositories, event lists, and placement instructions. To obtain knowledge, students have to go to different sites or rely on the teachers. Because of this decentralization, resource finding is ineffective and slow.



Fig 2. All in one dumped menu navigation

IV. PROPOSED METHODOLOGY

In order to (i) recommend skill-growth actions, (ii) match users with projects, mentors, and communities, (iii) provide a secure idea workspace with role-based visibility, and (iv) offer a startup-support chatbot to refine and operationalize ideas, Connect-X, a privacy-aware, AI-driven platform, was designed and implemented using the general methodology described in this section.

A. Design Goals and Assumptions

Goals

- unified experience: concentrating departmental collaboration on ideas, events, resources, and chat.
- Individualized advice: suggest next steps for skill development (projects, mentors, events, groups, resources).
- Role-based access control (RBAC), explicit consent, auditability, and explainability are all aspects of privacy-by-design.
- Fairness and inclusion: guarantee varied exposure across skill levels, cohorts, and departments.
- Real-time utility: live collaboration tools and prompt notifications.

Assumptions

- Through institutional SSO, users authenticate; their profiles contain their department, program year, and specified skills and interests
- Sensitive information is kept to a minimum; interaction logs (clicks, saves, joins, and comments) are first-party and stored with agreement.
- Event feeds and resource metadata are provided by departments; concept workspaces are by default private, however sharing is optional.



Fig 3. Subsystems in applications that are planned

B. System Overview and Data Flow

Architecture (high level).

- **Frontend:** HTML/CSS/JS single-page app.
- **Backend (Java):** REST APIs + WebSocket/SSE for real time; modules for **Matchmaking, Recommendation, Workspace, Events/Community, Chatbot, AuthN/AuthZ, and Notifications.**
- **Data stores:** Relational DB (e.g., PostgreSQL/MySQL) for entities and permissions; a search/embedding index for semantic retrieval.
- **Telemetry:** event bus for interaction events (view, click, save, apply, message).
- **Security:** OAuth2/OIDC SSO, RBAC checks at API gateway and service layer; encryption in transit and at rest.

Data Flow

1. Ingestion: metadata for projects, events, and resources; user profiles; skill tags; interactions; and optional course history.
2. Text normalization, tagging, embedding creation, and context attributes (time, department calendars) are examples of feature extraction.
3. Hybrid ranking (content + collaboration + context + fairness) is used for scoring.
4. Presentation: top-K matches and recommendations; clear justifications (important features/tags).
5. Feedback loop: models and exposure constraints are updated implicitly (clicks, dwell) and explicitly (thumbs/rating).

C. Data Model and Feature Engineering

Text features. Tokenized titles/descriptions; normalized tags; keyphrases (TF-IDF) and dense embeddings (e.g., sentence-level).

Profile/graph features. Skill vector s_u , interest vector i_u , activity vector a_u (recent interactions), department one-hot, year.

Item features. Tag vector t_c , difficulty, prerequisites, capacity, deadline/recency, department, modality (online/offline).

Context features. Semester phase, event proximity, user availability window, freshness decay.

Fairness features. Department diversity, cohort diversity, mentor load, exposure history.

D. Hybrid Recommendation & Matchmaking Pipeline

We employ a hybrid rating system that incorporates diversity/fairness requirements, contextual priors, collaborative filtering on implicit feedback, and content similarity.

a. Content Similarity

For user u with profile embedding $\mathbf{e}_u$ and candidate c (project/resource/event) with embedding $\mathbf{e}_c$:

$$\text{content_sim}(u,c) = \cos(\mathbf{e}_u, \mathbf{e}_c) \quad (1)$$

Embeddings come from pre-computed sentence encoders; for lightweight deployments, TF-IDF + SVD can be used.

b. Implicit Collaborative Filtering

Create an interaction matrix RRR and assign weights (e.g., 0.1/0.3/0.7/1.0/0.5) to views, clicks, saves, joins, and messages.

To estimate $\hat{r}_{u,c}$, train an implicit MF model (ALS) or a shallow model in the NCF style.

c. Contextual & Skill-Growth Signals

We associate next-step activities (e.g., "join a data-viz project," "complete a beginner lab in CV") with user intent (stated goals + recent activity). Let's:

$$\text{context_boost}(u,c) = \lambda_1 \text{freshness}(c) + \lambda_2 \text{deadline_proximity}(c) + \lambda_3 \text{intent_alignment}(u,c) \quad (2)$$

d. Diversity and Fairness (MMR-style)

To avoid popularity bias and to ensure cross-department exposure, we add a **diversity penalty** against items already in the slate S :

$$\text{diversity_adj}(u, c, S) = -\lambda_4 \max_{c' \in S} \cos(\mathbf{e}_c, \mathbf{e}_{c'}) \quad (3)$$

and a **fairness regularizer** to enforce exposure constraints across departments/cohorts:

$$\text{fairness_adj}(c) = \lambda_5 f(\text{dept}(c), \text{history}) \quad (4)$$

where f increases score for under-exposed segments.

e. Startup-Support Chatbot

goal. Assist students in developing, validating, and refining their business or project ideas; connect them with mentors or partners; and bring up events and references.

Problem → users → value prop → differential → risks → milestones is the idea scaffolding process.

Resource routing includes funding calls, clubs, labs, mentors, and auto-attach links to departmental or community resources.

Collaboration prompts: issue a call in pertinent forums (with consent) and suggest roles that are missing.

Explainable recommendations: the bot bases its recommendations on tags and current activities.

V. SYSTEM ARCHITECTURE

The presentation layer, application layer, AI & recommendation layer, and data management layer are the four main layers that make up Connect-X's modular, service-oriented architecture. Together, these layers provide chatbot-based startup support, secure collaboration tools, intelligent matchmaking, and real-time communication.

A. Presentation Layer

The user interface created using HTML, CSS, and JavaScript is provided by this layer. Through dashboards, idea workspaces, event views, chat modules, and the chatbot interface, it allows mentors, instructors, and students to engage with the platform. Using HTTPS, the user interface securely connects to backend APIs.

B. Application Layer (Backend)

RBAC and user authentication provide safe access. For versioning, private drafts, and controlled sharing, use the Idea Workspace Service.

For publishing announcements, conversations, and gatherings, use Events & Community Service. WebSocket-based notification service for updates, reminders, and real-time alerts.

C. AI & Recommendation Layer

The intelligence elements of Connect-X are powered by this layer:

Matchmaking Engine: recommends mentors, partners, and projects based on user interests, abilities, and activity logs.

Recommendation Engine: offers tailored recommendations for departmental opportunities, clubs, events, and educational materials.

Chatbot Engine: uses natural language processing (NLP)

and intent recognition to assist teamwork, skill-growth advice, and business ideation.

To produce pertinent results, hybrid models include contextual information, user behavior, diversity/fairness rules, and content similarity.

D. Data Management Layer

This layer stores and manages all structured and unstructured data:

- **Relational Database (MySQL/PostgreSQL):** user profiles, projects, events, permissions, interactions.
- **File/Object Storage:** idea documents, attachments, resource files.
- **Embedding/Feature Store:** vectors used for recommendation and semantic search.

Data is protected using encryption, access auditing, and strict role-based permissions.

E. Communication Layer

This layer manages:

- **API Gateway** for routing and securing all client requests.
- **Real-Time Channels (WebSockets/SSE)** for live updates, typing indicators, and collaboration signals.
- **External Integrations:** institutional SSO, email systems, department data feeds.

Supported by reliable databases and real-time communication layers, Connect-X is a multi-layered, AI-driven platform in which the frontend communicates with secure Java backend services that are linked to intelligent matchmaking and recommendation modules. Personalization, security, scalability, and smooth cross-departmental communication are all guaranteed by the design.

VI. CONCLUSION

Connect-X, an AI-powered, privacy-conscious departmental collaboration center that offers individualized, skill-growth assistance, was introduced in this study. A secure idea workspace with versioning and granular permissions, a context-aware discovery layer for events and departmental opportunities, a hybrid matchmaking and recommendation engine, and a startup-support chatbot that scaffolds ideation and connects students to mentors and collaborators. In terms of methodology, we described a modular architecture and a hybrid ranking strategy that combines diversity/fairness requirements, contextual boosts, implicit feedback, and content similarity. To guarantee responsible deployment at scale, we also described governance procedures (auditability, retention, and exposure monitoring) and evaluation measures (offline relevance, online engagement/conversion, fairness, and usability). Connect-X reduces coordination friction, speeds up team creation, and expands equal access to mentorship and resources across departments by combining collaboration, advice, and secure information sharing into a single platform. By doing this, it creates the foundation for an academic ecosystem that is sustainable and data-driven,

enabling students to move more quickly from concept to verified results, whether they be academic projects or startup endeavors. Deeper LMS/calendar linkages, more comprehensive on-device privacy models, mobile apps, and adaptive learning loops that further customize growth routes are examples of future additions.

REFERENCES

- [1] Pambudi, Ari, Ninda Lutfiani, Marviola Hardini, Achani Rahmania Az Zahra, and Untung Rahardja. "The digital revolution of startup matchmaking: Ai and computer science synergies." In 2023 Eighth International Conference on Informatics and Computing (ICIC), pp. 1-6. IEEE, 2023.
- [2] Jyotshna, Kuppili, and NP Lavanya Kumari. "CAREER GUIDANCE PLATFORM BASED ON AN AI CHATBOT AND RESUME GENERATOR."
- [3] Sitanggang, Christine. "Understanding Elderly Women's Needs for Building an AI-Powered Matchmaking Platform in Finland and Indonesia." (2023).
- [4] Faqihi, Ali, and Shah Jahan Miah. "Artificial intelligence-driven talent management system: Exploring the risks and options for constructing a theoretical foundation." *Journal of Risk and Financial Management* 16, no. 1 (2023): 31.
- [5] Shah, Priten. *AI and the Future of Education: Teaching in the Age of Artificial Intelligence*. John Wiley & Sons, 2023.
- [6] Islam, Aminul, Rifat Ali, Gursharan Singh, Bariul Islam, Aftabul Islam, and Sazzad Hossain. "An evaluation of AI-enhanced collaborative learning platforms." In 2024 International Conference on Communication, Computer Sciences and Engineering (IC3SE), pp. 207-211. IEEE, 2024.
- [7] Esangbedo, Caroline Olufunke, Jingxiao Zhang, Moses Olabhele Esangbedo, Seydou Dramane Kone, and Lin Xu. "The role of industry-academia collaboration in enhancing educational opportunities and outcomes under the digital driven Industry 4.0." *Journal of Infrastructure, Policy and Development* 8, no. 1 (2024): 2569.
- [8] Kovari, Attila. "A systematic review of AI-powered collaborative learning in higher education: Trends and outcomes from the last decade." *Social Sciences & Humanities Open* 11 (2025): 101335.
- [9] Scutelnicu, Liviu-Andrei, and Marius Ciprian Ceobanu. "Impact of AI-Powered Platforms on Academic Learning: Exploring the Purpose and Evaluation Used." In *Agents and Multi-agent Systems: Technologies and Applications 2024: Proceedings of 18th KES International Conference, KES-AMSTA 2024, June 2024, vol. 406, p. 371*. Springer Nature, 2025.
- [10] Vijayaraj, A., V. Sabapathi, Shai Kumar, R. Megavannan, and N. Mageshkumar. "Enhanced Intelligent Attendance Management System for Smart Campus Using Computer Vision." In 2025 2nd International Conference on Computing and Data Science (ICCDs), pp. 1-6. IEEE, 2025.
- [11] Sundarambal, Balaraman, N. Kirubakaran, Vishali Muralidharan, and M. Lokesh. "AI Based Departmental Management System." In 2025 Control Instrumentation System Conference (CISCON), pp. 1-11. IEEE, 2025.
- [12] Umamaheswari, K., S. S. Bhushan, K. S. Hariharan, G. Dennish Albert, and R. Harivarshan. "Revolutionizing Departmental Management through Intelligent Technology: A Machine Learning Approach to Optimization." In 2025 3rd International Conference on Artificial Intelligence and Machine Learning Applications Theme: Healthcare and Internet of Things (AIMLA), pp. 1-6. IEEE, 2025.
- [13] Lai, Qiuqia. "Research on the refinement of college student education management based on artificial intelligence." *Discover Artificial Intelligence* 6, no. 1 (2026): 31.
- [14] Saravanan, K., Nithish Kumar, and Shai Kumar. "A Gesture-based Mobile Platform for Accessible Education." In 2025 4th International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), pp. 410-415. IEEE, 2025.
- [15] Vijayaraj, A., T. Mahalingam, C. Teja Krishna, M. S. Sugumar, N. Vigneshkumar, and V. K. Yeshwanth. "Federated Hybrid AI-Powered Cybersecurity Framework for Adaptive Threat Detection." In 2025 2nd International Conference on Computing and Data Science (ICCDs), pp. 1-7. IEEE, 2025.

